# SE444: Introduction to Artificial Intelligence
# Mock Final Exam — Fall 2025

Prof. Anis Koubaa
akoubaa@alfaisal.edu
Alfaisal University

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## Instructions

- This is a closed-book exam. The University's code of ethics applies.
- Answer each exam question to the best of your knowledge.
- Show your work – any sign of serious effort will be considered.

## Student Information

| Student ID: | Student Name: |
|---|---|
| | |

**Summary of Exam Questions**

| Part | Topic | Time | Score | Mark | CLOs |
|---|---|---|---|---|---|
| **I MCQ** | BN Structure / d-Sep | 4m | 1 | | 1 |
| | Bayes' Rule | 4m | 1 | | 1 |
| | Decision Theory / MDPs | 4m | 1 | | 1,2 |
| | Game Theory | 4m | 1 | | 1,3 |
| | Planning (STRIPS) | 4m | 1 | | 1,2 |
| | Project Code | 4m | 1 | | 4 |
| **II BN** | BN + d-Separation | 12m | 4 | | 1 |
| | Bayesian Inference | 12m | 4 | | 2 |
| | MEU Decision | 9m | 3 | | 2,3 |
| **III MDP** | Value Iteration | 15m | 5 | | 2 |
| **IV Plan/GT** | STRIPS Planning | 9m | 3 | | 2 |
| | Nash Equilibrium | 9m | 3 | | 3 |
| **V Code** | Search Algorithms | 6m | 3 | | 4 |
| | Bayesian Updates | 6m | 2 | | 4 |
| | **TOTAL** | 102m | **30** | | |

## Part I. Critical-Thinking Multiple Choice Questions

**Total Score:** 6 points **Total Time:** 24 minutes

**Instructions:** Each question may have **one or more correct answers**. Select **all that apply**. Partial credit will be awarded for partially correct answers.

### Question I.1: Bayesian Network Structure & d-Separation

**Time:** 4 min **Score:** 1 pt **CLOs:** CLO 1

Consider the Bayesian Network: $A \to C \leftarrow B$, $C \to D$ (where $C$ is a collider).
Which conditional independence statements are TRUE?

A) $A \perp B$ (unconditionally)

B) $A \perp B \mid C$

C) $A \perp B \mid D$

D) $A \perp D \mid C$

E) $C \perp D$ (unconditionally)

### Question I.2: Bayes' Rule Application

**Time:** 4 min **Score:** 1 pt **CLOs:** CLO 1

A cybersecurity system: $P(\text{Intrusion}) = 0.01$, $P(\text{Alarm}|\text{Intrusion}) = 0.95$, $P(\text{Alarm}|\neg\text{Intrusion}) = 0.05$. The alarm triggers. Which are TRUE?

A) $P(\text{Intrusion}|\text{Alarm}) > 0.5$

B) $P(\text{Intrusion}|\text{Alarm}) < P(\text{Alarm}|\text{Intrusion})$

C) The base rate significantly affects the posterior

D) If $P(\text{Intrusion})$ increases to 0.10, then $P(\text{Intrusion}|\text{Alarm}) > 0.5$

E) Bayes' rule is unnecessary; $P(\text{Intrusion}|\text{Alarm}) = P(\text{Alarm}|\text{Intrusion})$

### Question I.3: Decision Theory & MDPs

**Time:** 4 min **Score:** 1 pt **CLOs:** CLO 1, 2

In an MDP with discount factor $\gamma = 0.9$, which statements are TRUE?

A) If $\gamma = 0$, the agent only considers immediate rewards

B) Bellman equation $V(s) = \max_a[R(s,a) + \gamma \sum_{s'} P(s'|s,a)V(s')]$ converges for $0 \leq \gamma < 1$

C) Value iteration and policy iteration produce the same optimal policy

D) Higher $\gamma$ (closer to 1) makes the agent more "farsighted"

E) In deterministic MDPs, optimal policy is the same regardless of $\gamma$

### Question I.4: Game Theory — Nash Equilibrium

**Time:** 4 min **Score:** 1 pt **CLOs:** CLO 1, 3

Consider the game (payoffs Row, Col):

| | **Left** | **Right** |
|---|---|---|
| **Up** | (3, 3) | (1, 4) |
| **Down** | (4, 1) | (2, 2) |

Which are TRUE?

A) Both players have a strictly dominant strategy

B) (Up, Left) is a Nash Equilibrium

C) (Down, Right) is a Nash Equilibrium

D) "Down" strictly dominates "Up" for Row player

E) The Nash Equilibrium yields lower payoffs for both players than (Up, Left)

### Question I.5: Classical Planning — STRIPS

**Time:** 4 min **Score:** 1 pt **CLOs:** CLO 1, 2

Consider the following STRIPS action schema:

```
Action: Load(cargo, truck, loc)
  Precond: At(cargo, loc), At(truck, loc), ¬In(cargo, truck)
  Add:     In(cargo, truck)
  Delete:  At(cargo, loc)
```

Which statements are TRUE?

A) Forward search starts from initial state and applies actions until goal

B) Closed World Assumption means unmentioned atoms are false

C) After Load(C1,T1,Depot), At(C1,Depot) is still true

D) Backward search can be more efficient when goal is specific

E) STRIPS assumes deterministic, fully observable effects

### Question I.6: Project Code — Search Algorithms

**Time:** 4 min **Score:** 1 pt **CLOs:** CLO 4

```
frontier = [(0, counter, start)]
while frontier:
    cost, _, current = heapq.heappop(frontier)
    if current in explored: continue
    explored.add(current)
    if current == goal: return path, cost
    for neighbor in env.get_neighbors(current):
        new_cost = cost + env.get_cost(current, neighbor)
        heapq.heappush(frontier, (new_cost, counter, neighbor))
```

Which are TRUE?

A) This implements Uniform Cost Search (UCS)

B) This implements A* Search with a heuristic

C) Goal test happens when node is **expanded** (popped)

D) With equal edge costs, this behaves like BFS

E) `counter` breaks ties in the priority queue

# Part II. Probability & Bayesian Networks

**Total Score:** 11 points **Total Time:** 33 minutes

## Question II.1: BN Construction & d-Separation (4 pts, 12 min)

**Scenario:** A company monitors network security with variables:

- `NetworkTraffic (N)`: High/Normal volume
- `IntrusionAttempt (I)`: Actual intrusion occurring
- `SystemLoad (S)`: Server under heavy load
- `FirewallAlert (F)`: Firewall generates alert
- `LogAnomaly (L)`: System logs show anomalies

**Dependencies:** $I$ and $S$ both cause high $N$. $F$ depends on $N$ and $I$. $L$ depends on $I$ and $S$.
**Tasks:**

(a) [2 pts] Draw the Bayesian Network structure.

(b) [1 pt] Are $I \perp S \mid F$? Use d-separation.

(c) [1 pt] Which node in this network is a **collider**? What happens when this collider node is observed?

## Question II.2: Bayesian Inference (4 pts, 12 min)

**Scenario:** A smart home system uses a motion sensor to automatically control lights. When someone is **Home**, the **Motion** sensor is more likely to detect movement. When **Motion** is detected, the **Lights** turn on automatically.

**Bayesian Network:** Home → Motion → Lights

| $P(\text{Home} = T)$ | 0.6 |
|---|---|

| $P(\text{Motion}\mid\text{Home})$ | Home$= T$ | Home$= F$ |
|---|---|---|
| Motion$= T$ | 0.9 | 0.1 |

| $P(\text{Lights}\mid\text{Motion})$ | Motion$= T$ | Motion$= F$ |
|---|---|---|
| Lights$= T$ | 0.95 | 0.05 |

**Tasks:**

(a) [2 pts] Given that the lights are ON, compute $P(\text{Home} = T \mid \text{Lights} = T)$ using enumeration.

(b) [2 pts] Apply Variable Elimination to compute $P(\text{Home} = T \mid \text{Lights} = T)$. Show factor tables.

## Question II.3: Decision Under Uncertainty (3 pts, 9 min)

**Scenario:** A drone delivery company must decide how to deliver a package. The drone's success and reward depend on the weather conditions, which are uncertain.

**Weather Probabilities:** Clear (0.6), Cloudy (0.3), Storm (0.1)

| **Action** | **Clear** | **Cloudy** | **Storm** |
|---|---|---|---|
| Fly Direct | +100 | +60 | −50 |
| Take Detour | +70 | +70 | +30 |
| Wait | +50 | +50 | +80 |

**Tasks:**

(a) [2 pts] Compute the **Expected Utility (EU)** for each action using: $EU(a) = \sum_s P(s) \cdot U(a, s)$

(b) [1 pt] Which action is optimal according to the **Maximum Expected Utility (MEU)** principle? Justify.

## Part III. MDPs & Value Iteration

**Total Score:** 5 points **Total Time:** 15 minutes

### Question III.1: MDP & Value Iteration (5 pts, 15 min)

**Scenario:** A warehouse robot needs to navigate a 2×3 grid to reach a goal position. The robot receives a reward when it reaches the goal, but incurs a small cost for each step taken.

**Grid Layout:**

| | | |
|---|---|---|
| S1 | S2 | S3 |
| S4 | S5 | **G** (+10) |

**Parameters:**

- **Goal reward:** $R(G) = +10$
- **Living reward (step cost):** $-1$ per step
- **Discount factor:** $\gamma = 0.9$
- **Transitions:** 80% intended direction, 10% slip left, 10% slip right. Walls keep robot in place.

**Tasks:**

(a) [1 pt] Write the **Markov Decision Process (MDP)** components: $(S, A, P, R, \gamma)$.

(b) [2 pts] Perform 2 iterations of **Value Iteration** for states S3 and S5. Start with $V_0 = 0$. Use the Bellman equation: $V_{k+1}(s) = \max_a [R(s, a) + \gamma \sum_{s'} P(s'|s, a) V_k(s')]$. Note: $V(G) = 10$.

(c) [2 pts] Extract the **optimal policy** $\pi^*(s)$ for S3 and S5. Justify your answer.

## Part IV. Planning & Game Theory

**Total Score:** 6 points **Total Time:** 18 minutes

### Question IV.1: Classical Planning — STRIPS (3 pts, 9 min)

**E-Commerce Fulfillment:** Predicates: `At(pkg,loc)`, `In(pkg,truck)`, `TruckAt(truck,dock)`, `Sorted(pkg)`.
**Initial:** At(P1,Intake), At(P2,Intake), TruckAt(T1,DockA), ¬Sorted(P1), ¬Sorted(P2)
**Goal:** In(P1,T1), In(P2,T1)
**Tasks:**

(a) [1 pt] Define STRIPS action `Load(pkg,truck,dock)`: Preconditions, Add List, Delete List.

(b) [2 pts] Show forward search plan to achieve goal. For each step: action applied, key state changes.

### Question IV.2: Game Theory — Nash Equilibrium (3 pts, 9 min)

**Tech Startup Competition:** Two startups choose pricing: Premium, Budget, or Freemium.

| A \ B | Premium | Budget | Freemium |
|---|---|---|---|
| **Premium** | (8, 8) | (3, 10) | (4, 9) |
| **Budget** | (10, 3) | (5, 5) | (6, 4) |
| **Freemium** | (9, 4) | (4, 6) | (7, 7) |

**Tasks:**

(a) [1 pt] Identify any strictly dominated strategies. Apply IESDS if applicable.

(b) [2 pts] Find all pure-strategy Nash Equilibria. Mark best responses for each player.

## Part V. Project Code Analysis

**Total Score:** 5 points **Total Time:** 12 minutes

### Question V.1: Search Algorithm Analysis (3 pts, 6 min)

```
def astar(env, start, goal, heuristic='manhattan'):
    h = lambda pos: env.manhattan_distance(pos, goal)
    g_score = {start: 0}
    f_score = {start: h(start)}
    frontier = [(f_score[start], counter, start)]

    while frontier:
        current_f, _, current = heapq.heappop(frontier)
        if current in explored: continue
        explored.add(current)
        if current == goal: return path, g_score[current]
        for neighbor in env.get_neighbors(current):
            tentative_g = g_score[current] + env.get_cost(current, neighbor)
            if neighbor not in g_score or tentative_g < g_score[neighbor]:
                g_score[neighbor] = tentative_g
                f_score[neighbor] = tentative_g + h(neighbor)
                heapq.heappush(frontier, (f_score[neighbor], counter, neighbor))
```

**Tasks:**

(a) [1 pt] What do `g_score`, `f_score`, and `h()` represent? Explain $f(n) = g(n) + h(n)$.

(b) [1 pt] Why check `if current in explored: continue`? What if removed?

(c) [1 pt] Is Manhattan distance admissible for 4-connected grid? Why/why not?

### Question V.2: Bayesian Belief Updates (2 pts, 6 min)

```
def update_single_cell_belief(prior, sensor_reading, sensor_accuracy=0.9):
    if sensor_reading:  # Obstacle detected
        likelihood_h = sensor_accuracy       # P(detect | obstacle)
        likelihood_not_h = 1 - sensor_accuracy  # P(detect | free)
```

```
    else:  # Clear
        likelihood_h = 1 - sensor_accuracy   # P(clear | obstacle)
        likelihood_not_h = sensor_accuracy   # P(clear | free)
    posterior, _ = bayes_update_full(prior, likelihood_h, likelihood_not_h)
    return posterior
```

**Tasks:**

(a) [1 pt] If prior=0.3, sensor_reading=True, accuracy=0.9, compute posterior. Show Bayes' rule.

(b) [1 pt] What happens if accuracy=0.5? Explain intuitively.